



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam, Trimester: Summer 2025

Course Code: CSE-3521 Course Title: Database Management Systems

Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

- 1 a) Consider the following relation and find out 4 valid Functional Dependencies from the given relation: 3

Employee_ID	Employee_Name	Department	Salary	Office_Location
101	Arif	HR	45000	Dhaka
102	Rafi	IT	55000	Chittagong
103	Nabila	Finance	60000	Dhaka
104	Arif	HR	48000	Khulna
105	Rafi	IT	53000	Chittagong

- b) A relation R(P, Q, R, S, T, U, V) is defined with the following dependencies:

$PQ \rightarrow R$

$PR \rightarrow S$

$S \rightarrow TU$

$T \rightarrow V$

$UV \rightarrow P$

- (i) **Determine** all candidate keys. 3
(ii) **Find** the minimal cover of the functional dependencies. 3
(iii) **Find** the set of non-prime attributes of R. 2
c) **Explain** the concept of dependency preservation in database decomposition. Why is it important when normalizing a relation? Support your explanation with a suitable example. 3

- 2 a) A company maintains a database table named **EMPLOYEE**: 3
EMPLOYEE (EmpID, Name, Department, Salary, JoiningDate)

Here, EmpID is unique for each employee. The HR department frequently runs queries to find all employees in a specific **Department** and occasionally, queries are made to search employees by **Salary range**. What type of indexing would be most suitable in this scenario?

- b) **Draw** a B+ tree of order 4 for the given key values below, 7
2, 9, 1, 7, 13, 19, 27, 14, 35, 37, 3

- 3 a) Consider the following hash function - 8

$$h(x) = x \% (\text{Global depth} * 2)$$

and the given search keys:

2, 7, 1, 24, 60, 28, 9, 14

Considering the LSB mapping of the hash values, insert the search keys one by one into the extensible hash structure using the given hash function. Assume that the bucket capacity is 2 and the initial local and global depths are 1.

- b) Does a hash structure work well for range queries (such as key BETWEEN 10 AND 20)? 2
Justify your answer.

- 4 a) Assume that the values of A and B stored in the database are 600 and 700. 3

T1	T2
Read(A) A=A-10; Read(C) Write (A)	
	Read(A) A=A+10; Write(A)
Write(C) Read(B) B=B+10; Write(B)	
	Read(B) Write(B)

Schedule X

Determine if schedule X is equivalent to a non-parallel schedule or not. If yes, produce the equivalent serial schedule by swapping instructions. If no, explain why not.

- b) In the following precedence graph given in figure 1, A, B, C,D and E are transactions. 3
Determine whether the schedule is conflict serializable or not (show the steps).

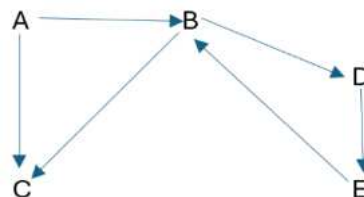


Fig: 1